

ECON 21030: Honors Econometrics

Homework 4

Spring Quarter 2024

Question 1: [5 points] Consider a regression of Y_i on a binary treatment D_i and a set of indicator variables W_i that span the support of a discrete control X_i . Assume we omit one element of W_i so that we can also include a constant. Show that the in following regression β recovers the average treatment effect assuming that $(Y(1), Y(0)) \perp\!\!\!\perp D_i | X_i$ and sufficient conditional variance in D_i given X_i for the parameters to be identified.

$$Y_i = \alpha + \beta D_i + W_i' \gamma_1 + D_i (W_i - E[W_i])' \gamma_2 + U_i$$

Question 2: [15 points] Consider a setting where $(Y(1), Y(0)) \perp\!\!\!\perp D_i | X_i$ again. Let $e(x) = Pr(D_i = 1 | X_i = x)$ be the propensity score. Assume that $e(x) \in (0, 1)$.

1. Show that the average treatment effect $E[Y_i(1) - Y_i(0)]$ is identified by:

$$E \left[\frac{Y_i D_i}{e(X_i)} \right] - E \left[\frac{Y_i (1 - D_i)}{1 - e(X_i)} \right]$$

2. Suppose I am instead interested in treatment on the treated. Propose a weighting scheme for $Y_i D_i$ and $Y_i (1 - D_i)$ that identifies this treatment effect from the weighted difference in means. Here, a weighted mean is just the expectation of the weights times the random variable divided by the sum of the weights.
3. Now suppose I am sure that $(Y(1), Y(0)) \perp\!\!\!\perp D_i | X_i$, but I don't know $e(X_i)$. Instead, I know that $E[Y_i | D_i = d, X_i] = \mu_d(X_i)$. Show that:

$$E[\mu_1(X_i) - \mu_0(X_i)]$$

identifies the average treatment effect.

4. Suppose that I *think* I know $e(X_i)$, but I'm not confident I have it right. I feel similarly about $\mu_d(X_i)$. Consider:

$$\beta_{DR} = E \left[\frac{D_i (Y_i - \mu_1(X_i))}{e(X_i)} + \mu_1(X_i) - \frac{(1 - D_i) (Y_i - \mu_0(X_i))}{1 - e(X_i)} - \mu_0(X_i) \right]$$

Under what conditions will β_{DR} be equal to the average treatment effects? Why might this be reassuring given what we do (and do not) know about $e(X_i)$ and $\mu_d(X_i)$?

5. Now consider the regression:

$$Y_i = \alpha + \beta D_i + X_i' \gamma_1 + D_i (X_i - E[X_i])' \gamma_2 + U_i$$

where X_i excludes a constant term. Suppose that X_i is not necessarily saturated—e.g., it includes a linear term for individual i 's age. When would it be wise to weight the regression by $W_i = \frac{D_i}{e(X_i)} + \frac{1-D_i}{1-e(X_i)}$?

As a reminder, regression weighting is covered at the end of our material on Gauss-Markov and GLS. A weighted estimate of a regression coefficient can be computed as:

$$\hat{\beta} = \left(\sum_{i=1}^N w_i X_i X_i' \right)^{-1} \left(\sum_{i=1}^N w_i X_i Y_i \right)$$

where w_i are the weights.

Hint: You may want to consider what the estimator of $\mu_d(X_i)$ implied by this regression is before you do any weighting.

- Now suppose that the regression is saturated, so that X_i contains a set of indicators for each point in the support of the controls, with one indicator omitted so that we can include the constant. Suppose that we have a random sample (Y_i, D_i, X_i) of size N . Show that the estimate of β from the regression:

$$Y_i = \alpha + \beta D_i + X_i' \gamma_1 + D_i (X_i - \bar{X})' \gamma_2 + U_i$$

is algebraically equivalent to the empirical propensity score estimate:

$$\hat{\beta}_{PS} = \sum_{i=1}^N \frac{Y_i D_i}{\hat{Pr}(D_i = 1 | X_i)} - \frac{Y_i (1 - D_i)}{1 - \hat{Pr}(D_i = 1 | X_i)}$$

where $\hat{Pr}(D_i = 1 | x) = \sum_{i=1}^N D_i 1\{X_i = x\} / \sum_{i=1}^N 1\{X_i = x\}$. Does it matter which one you estimator you use in this case?

Question 3: [10 points] Consider a setting where we have potential outcomes $Y_i(1)$ and $Y_i(0)$ indexed by a binary treatment D_i . Let X_i be a saturated set of controls partitioned into two subsets, Z_i and $W_i = X_i \setminus Z_i$, and assume an iid random sample (Y_i, D_i, X_i) is available.

- Suppose that $Pr(D_i = 1 | Z_i, W_i) = Pr(D_i | Z_i)$. Is controlling for both Z_i and W_i necessary to estimate a convex average of causal effects in the following regression?

$$Y_i = \beta D_i + Z_i' \gamma_1 + W_i' \gamma_2 + U_i$$

- Consider the homoskedastic estimator of the sampling variance of β . How is it affected by the inclusion of W_i ?
- What does this imply about the inclusion of extra control variables that are not necessary to estimate causal effects?

Question 4: [10 points] This problem asks you to conduct several empirical exercises based on the data from Kline, Rose, and Walters (2022, QJE). This paper studies a large-scale correspondence experiment where resumes with racially and gender distinctive names were sent to job vacancies. The researchers then recorded whether the employer contacted the job applicant for an interview within 30 days.

You can find the paper [here](#). And you can download the replication data at [this link](#).

1. Download the replication data and load the file *replication/data/data.dta* (this is Stata file) into the program of your choice. Each row in this dataset is an application sent to an employer. Construct an attractive table summarizing the total number of applications sent by race and gender category, along with any other characteristics you feel important.
2. Construct a table with mean contact rate by race and gender group along with appropriate standard errors. Report the p-value from a hypothesis test of the null that Black and White applications had the same contact rate. Also report a p-value from a hypothesis of the null that Male and Female applications had the same contact rate. Explain in the table notes how the p-values were constructed.
3. Run a single regression that reports the effect of application race and gender combinations on contact rates. Estimate appropriate standard errors and construct a table reporting the results. Make sure the table is clearly labeled. What do you conclude?
4. Construct a table with mean contact rate by first name for each race and gender combination. Also report standard errors on each contact rate. Summarize this in a graph with contact rates for each name (grouped by race / gender cell) and error bars given by ± 2 standard errors.
5. Conduct a hypothesis test for the null that each name's contact rate is the same within each race and gender combination. Report the four p-values from these tests, and explain how they were constructed. What do you conclude?